

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial 8 – methods and correlations for calculation of K_La - Solutions

Question 1

You were hired to work in a new biotechnology company that was founded as a daughter company of a European corporation. Upon entering to your new position, you had on your table results of an experiment done by another engineer in the company, when operating the fermenter on the first time. The experiment aimed to find the oxygen transfer coefficient of the fermenter during operation in the required conditions of the process, and was carried out using the titration method (Sulfite oxidation). The results of the experiment showed that the oxygen transfer rate in the fermenter is $67 \frac{\text{mmole } O_2}{\text{lit} \cdot \text{hr}}$.

From these experimental results you need to predict what will happen in the first fermentation that will be done in this fermenter. The fermentation will be continuous, with specific growth rate of 0.6 per hour and cells concentration of $2 \frac{\text{g}}{\text{lit}}$ during steady state. It is known that the cells yield on oxygen is $1200 \frac{\text{mgX}}{\text{gO}_2}$ and that the soluble oxygen concentration in saturation is $0.21 \frac{\text{mmole } O_2}{\text{lit}}$ at the given operation conditions.

- A. What is the oxygen transfer constant (K_La) at the given operation conditions?**
- B. What is the oxygen uptake rate (OUR) of the cells?**
- C. What is the soluble oxygen concentration in the medium during steady state?**
- D. There is an emergency electric generator installed in the factory, that is activated 20 second after power outage occur (which lead to mixing and aeration pause in the fermenter). By Murphy's law, In the first fermentation that you carried - the local power system collapsed (electrical corporation sanctions...) for a long while.**

Considering that the critical concentration of oxygen in this process is $0.021 \frac{\text{mmole } O_2}{\text{lit}}$, did the fermentation endured? What is your recommendation for future power outages?

Solution:

Given:

Titration:

$$OTR = 67 \frac{\text{mmole } O_2}{\text{lit} \cdot \text{hr}}$$

Continuous fermentation:

$$\mu = 0.6 \frac{1}{\text{hr}}$$

$$X_F = 2 \frac{\text{gX}}{\text{lit}}$$

$$Y_{\frac{X}{O}} = 1200 \frac{\text{mgX}}{\text{gO}_2} = 1.2 \frac{\text{gX}}{\text{gO}_2}$$

$$C^* = 0.21 \frac{\text{mmole } O_2}{\text{lit}}$$

A. $K_L a = ?$

In the Sulfite oxidation method $C_L = 0$, therefore:

$$OTR = K_L a \cdot (C^* - \cancel{C_L}) = K_L a \cdot C^*$$

$$K_L a = \frac{OTR}{C^*} = \frac{67 \frac{\text{mmole } O_2}{\text{lit} \cdot \text{hr}}}{0.21 \frac{\text{mmole } O_2}{\text{lit}}}$$

$$K_L a = 319 \frac{1}{\text{hr}}$$

B. $OUR = ?$

In standard continuous fermentation OUR is constant and equals to:

$$OUR = \frac{\mu \cdot X_F}{Y_{x/o}} = \frac{0.6 \frac{1}{hr} \cdot 2 \frac{gX}{lit}}{1.2 \frac{gX}{gO_2}}$$

$$OUR = 1 \frac{gO_2}{lit \cdot hr}$$

C. $C_L = ?$

$$OUR = OTR$$

$$OUR = K_L a \cdot (C^* - C_L)$$

$$C_L = C^* - \frac{OUR}{K_L a} = 0.21 \frac{mmole O_2}{lit} - \frac{1 \frac{gO_2}{lit \cdot hr} \cdot \frac{1 mole O_2}{32 gO_2} \cdot 10^3 \frac{mmole}{mole}}{319 \frac{1}{hr}}$$

$$C_L = 0.112 \frac{mmole O_2}{lit}$$

D. Given:

$$C_{Lmin} = 0.021 \frac{mmole O_2}{lit}$$

$$t = 20 sec$$

This is the first step in the dynamic method, when the aeration stops so that $OTR = 0$ and:

$$\frac{dC_L}{dt} = -OUR$$

After integration we get that:

$$\Delta C_L = -OUR \cdot \Delta t$$

We need to check how long it takes for the oxygen concentration in the fermenter to reach

C_{Lmin} :

$$C_{Lmin} - C_L = -OUR \cdot \Delta t$$

Since the time frame is very short, OUR does not change:

$$\Delta t = \frac{C_L - C_{L_{min}}}{OUR} = \frac{(0.112 - 0.021) \frac{mmole O_2}{lit}}{1 \frac{gO_2}{lit \cdot hr} \cdot \frac{1 mol O_2}{32 gO_2} \cdot 10^3 \frac{mmole}{mole}} = 2.9 \cdot 10^{-3} hr = 10.48 sec$$

The fermentation did not endure the power outage since $t = 20 sec > 10.48 sec$.

The recommendation would be to reduce the electric generator operation to 10 second.

Question 2

A factory performs batch growth of bacteria in 500-liter fermenter. At the beginning of the fermentation, the medium contains sugar in concentration of $70 \frac{g}{lit}$ ($Y_X = 0.52 \frac{gX}{gS}$). The bacteria are inoculated to the fermenter in initial concentration of $0.5 \frac{g}{lit}$ and immediately starts logarithmic growth. It can be assumed that the growth rate is constant until the end of the fermentation.

During the fermentation, at two different time points, the aeration was stopped, and the soluble oxygen concentration decline in the fermenter was tracked. The results of the experiment are described as follow: (1) after 2-hour fermentation the aeration was stopped when the oxygen electrode showed the value of 92.67%, and after 15 second the electrode reading declined to 40%. (2) after 8-hour fermentation the aeration was stopped when the oxygen electrode showed the value of 75.35%, and after 5 second the electrode reading declined to 15.45%. In both experiments described, the aeration was resumed immediately at the end of the experiment, and it can be assumed that the growth rate, as well as the biomass concentration, did not change during these experiments. The biomass yield on oxygen is $0.85 \frac{gX}{gO_2}$.

A. What is the oxygen transfer coefficient ($K_L a$) in the fermenter?

B. After how many hours will the fermentation end?

Solution:

Given:

Fermentation

$$V = 500 \text{ liter}$$

$$S_0 = 70 \frac{gS}{lit}$$

$$Y_{\frac{X}{S}} = 0.52 \frac{gX}{gS}$$

$$X_0 = 0.5 \frac{gX}{lit}$$

$$Y_{\frac{X}{O_2}} = 0.85 \frac{gX}{gO_2}$$

$$t = 2 \text{ hr}$$

$$\frac{C_L}{C^*} = 0.9267$$

$$\frac{C_L}{C_{t=15 \text{ sec}}^*} = 0.40$$

$$t = 8 \text{ hr}$$

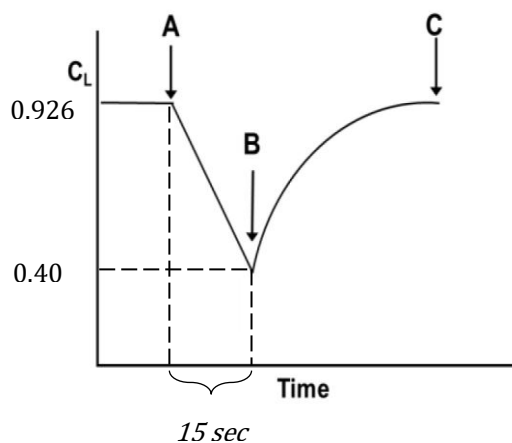
$$\frac{C_L}{C^*} = 0.7535$$

$$\frac{C_L}{C_{t=5 \text{ sec}}^*} = 0.1545$$

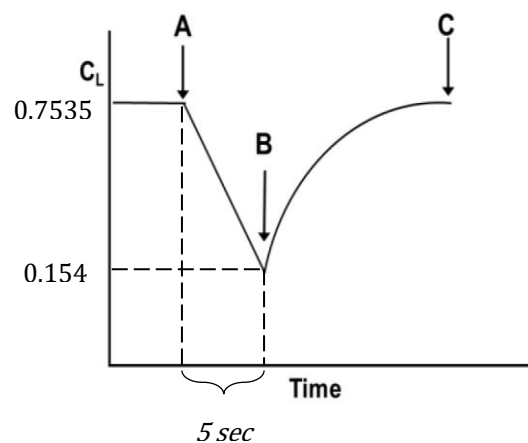
A. $K_L a = ?$

The experiment was done using the dynamic method:

First experiment after 2 hours



First experiment after 8 hours



We will calculate the average $K_L a$ from the results of the first and second experiments.

During aeration break $OTR = 0$, therefore:

$$\frac{dC_L}{dt} = -OUR$$

$$OUR (1) = -\frac{\Delta C_L}{\Delta t} = -\frac{(0.4 - 0.9267) \cdot C^*}{15 \text{ sec} \cdot \frac{1 \text{ hr}}{3600 \text{ sec}}} = 126.4 \cdot C^* \frac{1}{\text{hr}}$$

Since the experiment is very quick, OUR does not change and equals to the OTR before the experiment, therefore:

$$OUR = OTR = K_L a \cdot (C^* - C_{L_{t=2 \text{ hr}}})$$

$$K_L a(1) = \frac{OUR}{C^* - C_{L_{t=2 \text{ hr}}}} = \frac{126.4 \cdot C^* \frac{1}{\text{hr}}}{C^* - 0.9267 \cdot C^*} = 1724 \frac{1}{\text{hr}}$$

Do the same calculation with the data of the second experiment:

$$OUR (2) = -\frac{(0.1545 - 0.7535) \cdot C^*}{5 \text{ sec} \cdot \frac{1 \text{ hr}}{3600 \text{ sec}}} = 431.28 \cdot C^* \frac{1}{\text{hr}}$$

$$K_L a(2) = \frac{431.28 \cdot C^* \frac{1}{\text{hr}}}{C^* - 0.7535 \cdot C^*} = 1749 \frac{1}{\text{hr}}$$

Average the results:

$$K_L a(\text{avg.}) = 1736.5 \frac{1}{\text{hr}}$$

B. $t = ?$

The fermentation can be finished because of substrate or oxygen completion. We need to check which of these is potentially the limiting factor in the given case.

First, we will check what is the fermentation period (t_s) when the substrate is the limiting factor, so $S_F = 0$.

We can find t_S from the growth equation:

$$\ln\left(\frac{X_F}{X_0}\right) = \mu \cdot t$$

We have X_0 , so we need to calculate μ and X_F .

X_F can be found from the yield $Y_{X/\frac{S}{S}}$:

$$Y_{X/\frac{S}{S}} = \frac{X_F - X_0}{S_0 - \cancel{S_F}}$$

$$X_F = Y_{X/\frac{S}{S}} \cdot S_0 + X_0 = 0.52 \frac{gX}{gS} \cdot 70 \frac{gS}{lit} + 0.5 \frac{gX}{lit} = 36.9 \frac{gX}{lit}$$

μ can be found from OUR :

For $t = 2 \text{ hr}$:

$$OUR_{t=2 \text{ hr}} = \frac{\mu \cdot X_{t=2 \text{ hr}}}{Y_{X/\frac{S}{S}}} = 126.4 \cdot C^* \frac{1}{hr}$$

For $t = 8 \text{ hr}$:

$$OUR_{t=8 \text{ hr}} = \frac{\mu \cdot X_{t=8 \text{ hr}}}{Y_{X/\frac{S}{S}}} = 431.8 \cdot C^* \frac{1}{hr}$$

It is given that we can assume that μ is constant, so we can divide the equations above:

$$\frac{OUR_{t=8 \text{ hr}}}{OUR_{t=2 \text{ hr}}} = \frac{431.8 \cdot C^* \frac{1}{hr}}{126.4 \cdot C^* \frac{1}{hr}} = 3.41$$

Now we can use the growth equation for the difference between these time points:

$$\ln\left(\frac{X_{t=8 \text{ hr}}}{X_{t=2 \text{ hr}}}\right) = \mu \cdot 6 \text{ hr}$$

$$\mu = \frac{\ln(3.41)}{6 \text{ hr}} = 0.204 \frac{1}{hr}$$

Last, we can calculate the fermentation time:

$$t = \frac{\ln\left(\frac{X_F}{X_0}\right)}{\mu} = \frac{\ln\left(\frac{36.9 \frac{gX}{lit}}{0.5 \frac{gX}{lit}}\right)}{0.204 \frac{1}{hr}} = 21 \text{ hr}$$

Second, we will check what is the fermentation period (t_o) when the oxygen is the limiting factor.

At the end of the fermentation: $OUR_{max} = OTR_{max}$

$$K_L a (C^* - \cancel{C_{limite}}) = \frac{\mu \cdot X_F}{Y_{X/O}}$$

$$X_F = \frac{K_L a \cdot C^* \cdot Y_{X/O}}{\mu}$$

We have all the parameters beside C^* . We will use the equation we found previously to find it:

$$\frac{\mu \cdot X_{t=2 \text{ hr}}}{Y_{X/O}} = 126.4 \cdot C^* \frac{1}{hr}$$

$$C^* = \frac{\mu \cdot X_{t=2 \text{ hr}}}{Y_{X/O} \cdot 126.4 \frac{1}{hr}}$$

$X_{t=2 \text{ hr}}$ can be found from the growth equation:

$$\ln \frac{X_{t=2 \text{ hr}}}{X_0} = \mu \cdot 2 \text{ hr}$$

$$\ln \frac{X_{t=2 \text{ hr}}}{0.5 \frac{gX}{lit}} = 0.204 \frac{1}{hr} \cdot 2 \text{ hr} \rightarrow X_{t=2 \text{ hr}} = 0.75 \frac{gX}{lit}$$

Back to the equation to find C^* :

$$C^* = \frac{0.204 \frac{1}{hr} \cdot 0.75 \frac{gX}{lit}}{0.85 \frac{gX}{gO_2} \cdot 126.4 \frac{1}{hr}} = 1.4 \cdot 10^{-3} \frac{gO_2}{lit}$$

Back to the equation to find X_F :

$$X_F = \frac{1736.5 \frac{1}{hr} \cdot 1.4 \cdot 10^{-3} \frac{gO_2}{lit} \cdot 0.85 \frac{gX}{gO_2}}{0.204 \frac{1}{hr}} = 10.13 \frac{gX}{lit}$$

We can already see that in this case (of oxygen as limiting factor) X_F is smaller, so we can already tell that the oxygen is the limiting factor. The fermentation period will be:

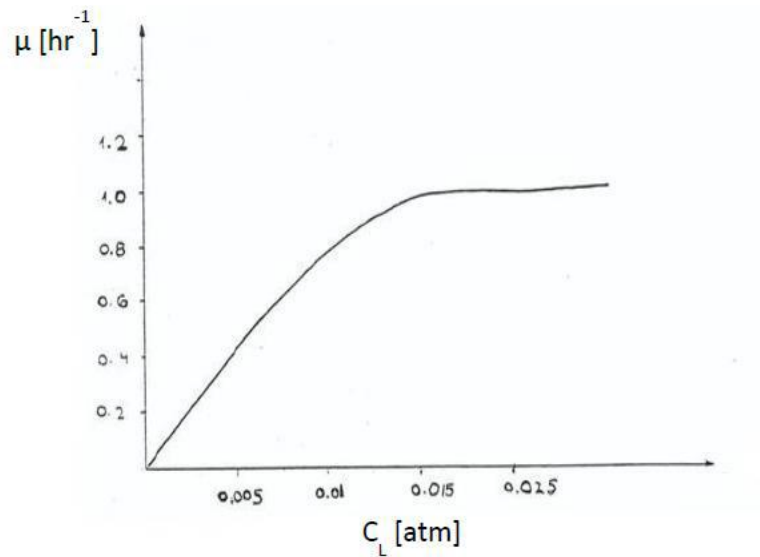
$$t = \frac{\ln\left(\frac{X_F}{X_0}\right)}{\mu} = \frac{\ln\left(\frac{10.13 \frac{gX}{lit}}{0.5 \frac{gX}{lit}}\right)}{0.204 \frac{1}{hr}} = 14.75 \text{ hr}$$

Question 3

Congratulations! You were graduated your studies in the biotechnology and food engineering department at GTIIT, and now you are going to an interview in a dynamic biotechnology factory. This factory aims to produce restriction enzymes, so it bought a number of secondhand fermenters from Shantou university.

The factory head biologist is trying to impress, and tells you that the fermenter purchasing was a complete nonsense because they can't supply enough oxygen to cell cultures. Because you also want to impress your new supervisors, you request information on the fermenters and the bacteria growth conditions, and this is what you get: the fermenters geometrical volume is 3 liter while the liquid volume in it is 2 liters. The proportion coefficient is $K=42$, the diameter of the tank is 25 cm and the diameter of the impeller is 6.7 cm. The mixing speed can be between 0 to 1000 rpm and the engine have maximal power of $5 \cdot 10^7 \frac{gr \cdot cm^2}{sec^3}$.

Simple aquarium pumps supply air to the fermenters in rate of $50 \frac{ml_{air}}{sec}$. It is known as well that the bacteria concentration can reach up to $2 \frac{gX}{lit}$ and the bacteria yield on oxygen is $1.6 \frac{gX}{gO_2}$. The biologist of the factory also gives you a graph from the literature which describes the bacteria growth rate as a function of oxygen concentration at 37° C .



Do you agree with the head biologist that the fermenter purchasing was a mistake?

Solution:

Given:

$$V_T = 3 \text{ lit}$$

$$V_L = 2 \text{ lit}$$

$$K = 42$$

$$D_T = 25 \text{ cm}$$

$$D_i = 6.7 \text{ cm}$$

$$N_{max} = 1000 \text{ rpm}$$

$$P = 5 \cdot 10^7 \frac{\text{gr} \cdot \text{cm}^2}{\text{sec}^3}$$

$$Q_{air} = 50 \frac{\text{ml}}{\text{sec}}$$

$$X_{max} = 2 \frac{\text{gX}}{\text{lit}}$$

$$Y_{\frac{X}{O}} = 1.6 \frac{\text{gX}}{\text{gO}_2}$$

We need to find out whether the fermenter can supply enough oxygen to the cells. We can check what is the maximal oxygen uptake in the system:

$$OUR_{max} = \frac{\mu_{max} \cdot X_{max}}{Y_{\frac{X}{O}}}$$

From the graph: $\mu_{max} = 1 \frac{1}{hr}$, therefore:

$$OUR_{max} = \frac{1 \frac{1}{hr} \cdot 2 \frac{gX}{lit}}{1.6 \frac{gX}{gO_2}} = 1.25 \frac{gO_2}{lit \cdot hr}$$

Let's check if the system can handle this:

$$OUR_{max} = OTR_{max}$$

$$OTR_{max} = K_L a \cdot (C^* - C_{Lmin})$$

From the graph: $C_{Lmin} = 0.015 atm$ (C_L when the growth rate starts to decline).

We will assume that C^* equals to C^* in water at 37°C: $C^* = 0.21 atm$.

Now we can find $K_L a$ by correlations:

For small fermenter:

$$K_L a = K \cdot \left(\frac{P_g}{V_L} \right)^{0.95} \cdot V_S^{0.67}$$

The linear air velocity V_S :

$$V_S = \frac{Q}{A} = \frac{50 \frac{ml}{sec} \cdot 60 \frac{sec}{min}}{\pi \cdot \frac{(25 cm)^2}{4}} = 6.1 \frac{cm}{min}$$

The power in aired system (P_g):

We assume Newtonian liquid and use the correlation for fermenters up to 4200 lit:

$$P_g = 25 \cdot 10^{-3} \cdot \left[\frac{P^2 \cdot N \cdot D_i^3}{Q^{0.56}} \right]^{0.45}$$

P and P_g in unit of HP

N in rpm

D_i in cm

Q in $\frac{\text{lit}}{\text{min}}$

We need to convert some unit for given parameters:

$$P = 5 \cdot 10^7 \frac{\text{gr} \cdot \text{cm}^2}{\text{sec}^3} \cdot 10^{-7} \frac{W}{\frac{\text{gr} \cdot \text{cm}^2}{\text{sec}^3}} \cdot \frac{1 \text{ HP}}{745.7 W} = 6.7 \cdot 10^{-3} \text{ HP}$$

$$Q_{\text{air}} = 50 \frac{\text{ml}}{\text{sec}} \cdot 60 \frac{\text{min}}{\text{sec}} \cdot 10^{-3} \frac{\text{lit}}{\text{ml}} = 3 \frac{\text{lit}}{\text{min}}$$

$$P_g = 25 \cdot 10^{-3} \cdot \left[\frac{(6.7 \cdot 10^{-3})^2 \cdot 1000 \cdot 6.7^3}{3^{0.56}} \right]^{0.45} = 6.11 \cdot 10^{-3} \text{ HP}$$

$$\frac{P_g}{V_L} = \frac{6.11 \cdot 10^{-3} \text{ HP}}{\frac{2 \text{ lit}}{1000 \text{ lit}} \cdot 1000 \text{ lit}} = 3.06 \frac{\text{HP}}{1000 \text{ lit}}$$

$$K_L a = 42 \cdot (3.06)^{0.95} \cdot 6.1^{0.67} = 408.2 \frac{\text{mmol } O_2}{\text{lit} \cdot \text{hr} \cdot \text{atm}}$$

$$OTR_{\text{max}} = 408.2 \frac{\text{mmol } O_2}{\text{lit} \cdot \text{hr} \cdot \text{atm}} (0.21 - 0.0015) \text{ atm} = 79.6 \frac{\text{mmol } O_2}{\text{lit} \cdot \text{hr}} = 2.55 \frac{gO_2}{\text{lit} \cdot \text{hr}}$$

$$OTR_{\text{max}} = 2.55 \frac{gO_2}{\text{lit} \cdot \text{hr}} > OUR_{\text{max}} = 1.25 \frac{gO_2}{\text{lit} \cdot \text{hr}}$$

Therefore, the fermenter can meet the requirements.

Question 4

Fermenter of 200 liter is used to grow *bacillus subtilis* culture at a temperature of 28°C. The air exits the fermenter at a rate of $189 \frac{\text{lit}}{\text{min}}$ with 20.1% oxygen. The electrode measures the soluble oxygen concentration shows that the liquid inside the fermenter is 52% saturated. The oxygen saturation solubility in the medium in the given temperature is $7.8 \frac{\text{mg } O_2}{\text{lit}}$.

A. Calculate the oxygen transfer rate (OTR).

B. What is $K_L a$ ($\frac{1}{\text{hr}}$)?

Solution:

Given:

$$V = 200 \text{ lit}$$

$$T = 28^\circ\text{C} = 301^\circ\text{K}$$

$$Q_{out} = 189 \frac{\text{lit}}{\text{min}}$$

$$\%O_{2_{out}} = 20.1\%$$

$$\frac{C_L}{C^*} = 0.52$$

$$C^* = 7.8 \frac{\text{mg } O_2}{\text{lit}}$$

A. $OTR = ?$

Balance of the oxygen in the air:

$$OTR = O_{2in} - O_{2out}$$

We assume that:

$$P = 1 \text{ atm} ; \%O_2 = 21\% ; Q_{in} = Q_{out}$$

$$OTR = 189 \frac{\text{lit}}{\text{min}} \cdot \frac{1 \text{ atm}}{0.082 \frac{\text{lit} \cdot \text{atm}}{\text{mole} \cdot ^\circ K} \cdot 301^\circ K} \cdot (0.21 - 0.201) \frac{\text{lit}_{O_2}}{\text{lit}_{air}} = 0.069 \frac{\text{mole}}{\text{min}}$$

Convert units:

$$OTR = 0.069 \frac{\text{mole}}{\text{min}} \cdot 32 \frac{\text{g}}{\text{mole}} \cdot 60 \frac{\text{min}}{\text{hr}} \cdot \frac{1}{200 \text{ lit}}$$

$$OTR = 0.66 \frac{\text{g}O_2}{\text{lit} \cdot \text{hr}}$$

B. $K_L a = ?$

$$OTR = K_L a \cdot (C^* - C_L)$$

$$K_L a = \frac{OTR}{(C^* - C_L)} = \frac{0.66 \frac{\text{g}O_2}{\text{lit} \cdot \text{hr}}}{7.8 \frac{\text{mg } O_2}{\text{lit}} (1 - 0.52) \cdot 10^{-3} \frac{\text{g}}{\text{mg}}}$$

$$K_L a = 176.3 \frac{1}{\text{hr}}$$